

Samuel Chua

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EDUCATION

University of Pennsylvania

Master of Science in Engineering in Robotics

Philadelphia, PA

Expected Graduation: May 2027

- GPA: 3.9/4.0
- Relevant coursework: Machine Perception, Optimization Theory, State Estimation, Neural Scene Representation and Rendering, Reinforcement Learning, Linear Control, Autonomous Racing

University of California, Los Angeles

Bachelor of Science in Computer Science & Engineering

Los Angeles, CA

Jun 2025

- GPA: 3.8/4.0
- Upsilon Pi Epsilon - Officer (Computing Honor Society), Eta Kappa Nu - Officer (ECE Honor Society)
- Relevant coursework: Data Structures & Algorithms, Discrete Structures, Computer Architecture, Software Construction, Algorithms & Complexity, Feedback Control, Deep Learning, Operating Systems, State Estimation

TECHNICAL SKILLS

Languages: Python, C++, C, Javascript, Java, HTML/CSS, Matlab, Verilog

Frameworks: React Native, Django REST, Tensorflow, PyTorch, ROS2, Express, Next.js, Arduino, GNURadio

Libraries: React, Scikit-learn, Ardupilot, Matplotlib, Numpy, HuggingFace, Detectron2, fast.ai

Others: Linux/Unix, Git, Postman, Eagle, PCB Design

EXPERIENCE

Kumar Lab @ GRASP Lab, Philadelphia, PA | *Graduate Research Assistant*

Oct 2025 – Present

- Developed semantic autonomy framework bridging metric-semantic mapping and planner-ready graph abstractions
- Designed semantic-topological graph representations, encoding semantic relevance, traversability, and connectivity
- Built a closed-loop language-guided exploration pipeline using foundation models for autonomous navigation

Linkedin, Mountain View, CA | *AI Infrastructure Intern*

May 2026 – Aug 2026

- Built deployment infrastructure supporting GitHub-less AI model deployments for internal search platform
- Automated validation of manifests into Kubernetes specifications for GPU and CPU inference workloads

VECTR Lab @ UCLA, Los Angeles, CA | *Robotics Researcher*

Dec 2023 – Jun 2025

- Developed online geometric change detection framework for autonomous robots in dynamic environments
- Designed scene decomposition and dense submap generation methods to efficiently detect local changes
- Achieved ~ 90% agreement with state-of-the-art offline methods, while enabling real-time map reconstruction

DSO National Laboratories, Singapore | *Robotics Algorithm Intern*

Jul 2024 – Sep 2024

- Designed degeneracy-aware LIDAR-Inertial Odometry (LIO) algorithm for high-speed robot autonomy
- Integrated Direct LIO and degeneracy detection for robust navigation in feature-sparse, dynamic environments
- Field-tested algorithms on challenging terrain, improving the reliability of legged robots in real-world conditions

DSO National Laboratories, Singapore | *Robotics Algorithm Developer*

Aug 2023 – Sep 2023

- Developed Viewpoint Generation algorithm for 3D urban search, optimizing inspection for swarm robotics drones
- Built a C++ quad-tree for dynamic surface subdivision, significantly boosting inspection efficiency by 70%
- Ensured complete surface coverage by validating drone camera footprints and checking for occlusion

LEMUR Lab @ UCLA, Los Angeles, CA | *Student Researcher*

Nov 2022 – Dec 2023

- Formulated a robust Deep Reinforcement Learning algorithm to optimize distribution of large-scale robot systems
- Employed decentralized controllers in Python to achieve Nash Equilibrium, enhancing Uber dispatching operations
- Publication: 3rd Author on "Population-aware Online Mirror Descent for Mean-Field Games by Deep Reinforcement Learning" International Conference on Autonomous Agents and Multiagent Systems

PROJECTS

Pill.ai (LA Hacks 2024 Winner) | github.com/SamuelChua/pill.ai

Apr 2024 – Apr 2024

- Engineered an LLM-driven web-app with React JS & Material UI, reducing medication-related errors by 90%
- Leveraged Google's Gemini API, utilizing 1M tokens for precise speech-to-text and image-to-text translation
- Successfully deployed a user-friendly and real-time dashboard for medication verification, benefiting 500+ patients